

Demo: Trivana: On-Device Multimodal Forecasting and LLM-Driven Mental Health Support

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Abstract—We present Trivana, an on-device mental health mobile application that enables holistic, proactive, and privacy-preserving support. The system integrates multimodal user inputs, a forecasting model for anticipating mental health risks, and a local LLM-based chatbot for personalized interventions. By unifying sensing, forecasting, and interaction within a single pipeline, Trivana overcomes fragmentation in existing apps while ensuring privacy guarantees through fully on-device execution.

Index Terms—Mental Health, Smartphone Sensing, LLM

I. INTRODUCTION

Mental health challenges are increasingly prevalent, while access to timely and affordable care remains limited due to high cost, limited availability of professionals, and social stigma [1]. Mobile mental health apps have emerged as a promising solution, offering low-cost, always-available, and scalable support seamlessly integrated into daily lives.

However, existing mental health apps exhibit three key limitations. First, they provide fragmented functionality across categories such as screen-time restriction tools, conversational agents, mindfulness training programs, and habit-building apps, which lacks a holistic view of the user’s mental and behavioral state. Second, most apps focus on detecting distress after it has already manifested or require users to initiate support. This is not desirable as individuals are often least motivated to seek help during periods of distress, thereby placing an additional burden on the user. Third, most large language model (LLM) based conversational systems (e.g., chatbot) rely on cloud-based infrastructure that requires user data to be shared with external providers, raising privacy concerns for sensitive mental health data.

To address these limitations, we present Trivana, an Android-based mental health app that enables holistic understanding, proactive care, and on-device LLM interaction. It leverages multimodal inputs (e.g., phone usage, activity) to capture user behavior across diverse contextual dimensions, thereby supporting a holistic view of mental well-being for personalized solutions. The system further incorporates state-of-the-art forecasting models [2] for preventative care. Finally, the entire system, including the LLM-based chatbot, runs on device to preserve user privacy.

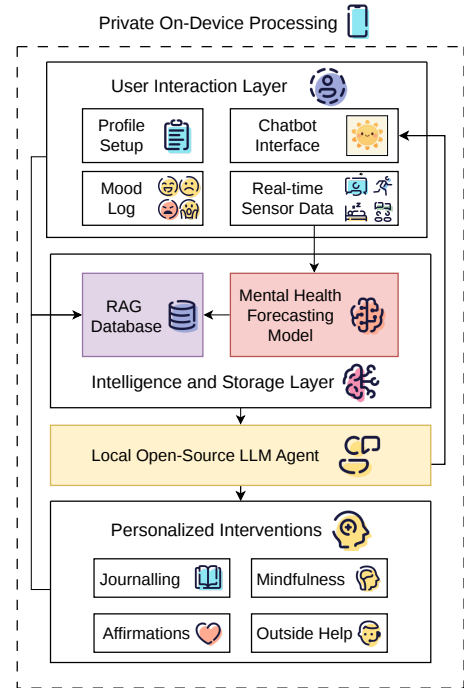


Fig. 1: The workflow of the Trivana mobile application.

II. SYSTEM OVERVIEW

A. Trivana Design

Trivana is the first known privacy-preserving, on-device mental health support system that integrates sensing, forecasting, and personalized intervention in a unified pipeline. As shown in Figure 1, the workflow begins with the user interaction layer, where users provide explicit inputs (e.g., profile setup and mood logs) and implicit signals through real-time sensor data. These heterogeneous data streams are processed locally and stored within an intelligence and storage layer, which includes a retrieval-augmented generation (RAG) database to maintain personalized history. At the core of Trivana is a mental health forecasting model that continuously analyzes multimodal inputs to predict mental health states,

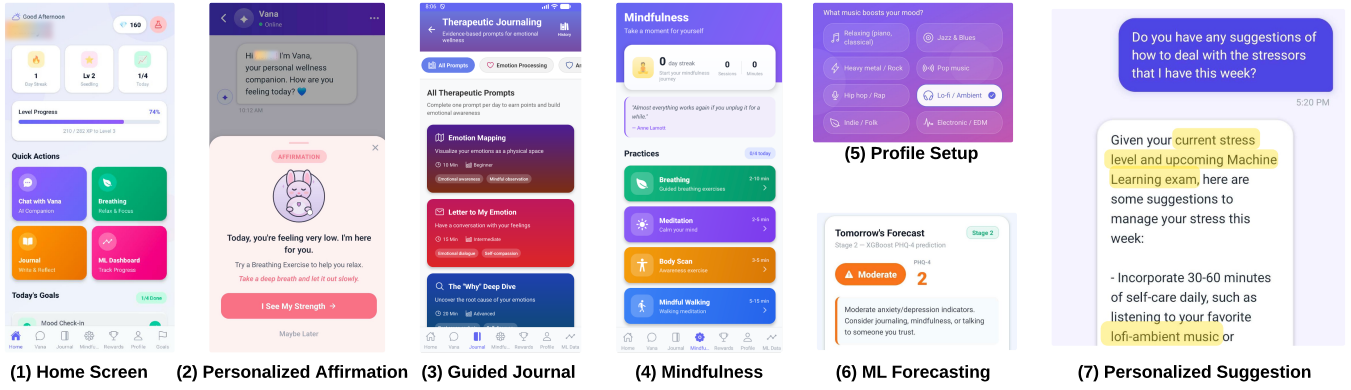


Fig. 2: Screenshots of core features in Trivana.

enabling proactive rather than reactive support. Building on these predictions, a local open-source LLM agent serves as an interactive chatbot interface that grounds its responses in both the RAG database and model outputs, allowing context-aware, personalized, and privacy-preserving interactions. Finally, the system delivers tailored interventions—such as journaling prompts, mindfulness exercises, or recommendations for help—closing the loop between sensing, prediction, and action. This tight integration of on-device forecasting and LLM-driven reasoning enables Trivana to provide real-time mental health support while maintaining user data privacy.

B. Core Features

Trivana builds on three core features that enable continuous, personalized, and privacy-preserving mental health support.

Multimodal Inputs. Trivana integrates passive and self-reported inputs to build a holistic view of mental well-being. As shown in Figure 2(1-5), users provide various self-reported inputs, while passive sensing signals are continuously collected via on-device sensors (e.g., activity, device usage). These inputs are complemented by structured interactions such as guided journaling (Figure 2(3)) and mindfulness engagement (Figure 2(4)), which not only support interventions but also serve as additional behavioral signals.

Mental Health Forecasting. Trivana includes a forecasting model that transforms multimodal inputs into short-term mental health forecasts. As shown in Figure 2(6), it produces interpretable predictions (e.g., PHQ-4-based risk level Moderate [3]) with actionable context, enabling proactive intervention by anticipating near-future mental states from historical temporal behavioral patterns. The model is designed to operate entirely on-device, ensuring low-latency inference and preserving user privacy, while supporting continuous updates as new data becomes available.

On-Device LLM Engine. Trivana incorporates an on-device LLM agent for conversational interaction and personalized interventions. As shown in Figure 2(2)(7), it generates context-aware responses by grounding in forecasting outputs and user history (via RAG), enabling dynamic guidance while preserving privacy through fully local processing. This design

not only enhances privacy and trust but also allows tightly coupled interaction between prediction and intervention, forming a closed-loop system for real-time mental health support.

C. Implementation Details

Trivana is implemented as a React Native mobile application, with the frontend and all on-device services developed in TypeScript and JavaScript. The backend is built in JavaScript using Node.js and Express.js, exposing RESTful APIs for the frontend. The machine learning forecasting component is a two-stage hierarchical model, which uses XGBoost as the base learner. The design builds upon prior work [2], [4]. The model is implemented in Python using FastAPI and scikit-learn. The on-device LLM-based chatbot leverages a quantized Qwen2.5-3B-Instruct model (Q4_K_M) executed via llama.rn (llama.cpp bindings for React Native). Retrieval-augmented generation (RAG) is supported through all-MiniLM-L6-v2 (384-dimensional) embeddings via @xenova/transformers (ONNX), with a TF-IDF fallback mechanism. Data storage is managed locally using an in-memory layer backed by SQLite through react-native-sqlite-storage, enabling fully on-device operation. By enabling proactive support and privacy-preserving operation, Trivana demonstrates a practical path toward scalable, personalized mental health care.

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